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1. BedMed Protocol (version date May 2, 2016), as submitted to the University of Alberta's Health Research Ethics Board
2. Log of all protocol amendments

*Note: The final version of the BedMed protocol, along with the final analysis plan, was published on Feb 24, 2022 as: *Garrison SR, Kolber MR, Allan GM, Bakal J, Green L, Singer A, Trueman DR, McAlister FA, Padwal RS, Hill MD, Manns B, McGrail K, O'Neill B, Greiver M, Froentjes LS, Manca DP, Mangin D, Wong ST, MacLean C, Kirkwood JE, McCracken R, McCormack JP, Norris C, Korownyk T. Bedtime versus morning use of antihypertensives for cardiovascular risk reduction (BedMed): protocol for a prospective, randomised, open-label, blinded end-point pragmatic trial. BMJ Open. 2022 Feb 24;12(2):e059711. doi: 10.1136/bmjopen-2021-059711. PMID: 35210352; PMCID: PMC8883279.*

BedMed Protocol

Problem Statement: In Canada cardiovascular disease (CVD), most notably heart attack (MI), stroke and congestive heart failure (CHF) accounts for 1 in 3 deaths, and 1 in 6 hospitalizations.¹ Hypertension, a modifiable risk factor, predisposes patients to all types of CVD and 19% of adult Albertans are hypertensive.² Using population-based Alberta administrative data (2002 to 2010) and using a regression model that accounts for comorbidities and socioeconomic factors, it was recently estimated that Alberta healthcare costs attributable to hypertension were \$1.4 billion in 2010.³ If high blood pressure can be reduced in a way that is more effective at preventing these life threatening and potentially disabling events, better health outcomes and substantial cost savings are possible.

Background: Blood pressure (BP) normally exhibits a circadian rhythm with relatively lower pressures during sleep.⁴ Lack of this sleep time “dip” correlates strongly with adverse cardiovascular events and BP correlates most strongly with such events when measured at night (i.e. during sleep).⁵⁻⁸ Motivated by such observations, Spanish researchers studied the effect of taking BP medication at BEDTIME (when the effect on nighttime BP would be greatest) versus conventional morning use, when most BP medications (meds) are taken. The results of this study (the MAPEC trial) were striking.⁹ Over a median 5.6 years follow-up, adverse cardiovascular events occurred in **187 of 1084** subjects taking BP medication in the morning but **only 68 of 1072** subjects who took their BP medication at bedtime (relative risk 0.39, 95%CI [0.29-0.51], $p < 0.001$). This 61% reduction in adverse events was similar for all individual components of the primary outcome (including death from all causes, stroke, MI, new angina pectoris, CHF and occlusion of the retinal artery). If true, a switch to bedtime prescribing would have more impact on the health of hypertensive Albertans than whether high BP is treated at all.

And yet this trial failed to change practice. This is because 1) most physicians are unaware of the findings; and 2) **replication of such surprising results is required for these findings to be believed** – primarily because the benefit seems “too good to be true”, far exceeding all other methods of CVD risk reduction (e.g. high dose statins convey a 35% relative risk reduction, morning BP meds 30%, exercise 25%, aspirin 15%).¹⁰ Neither a relatively recent (Oct 2013) literature review, nor a 2011 Cochrane systematic review found any RCTs (besides MAPEC) evaluating the effect of BP medication timing on “hard” endpoints.^{11,12} *The only other published trials looking at the timing of BP medication examine its effect on blood pressure – not on cardiovascular events.* The National Institute of Health recently funded a planning grant for a trial looking at patient oriented outcomes (e.g. stroke, MI, CHF) but that trial has yet to be registered.¹¹

Objectives:

- A) To determine if switching one or more BP meds to bedtime alters important clinical outcomes.
- B) To identify populations (e.g. frail elderly) that might be put at risk by bedtime administration.
- C) To determine whether diuretics can be successfully taken at night without excessive urination (this objective is in response to questions raised during our patient engagement).
- D) To determine differences in acute care costs between AM and bedtime antihypertensive users.
- E) To disseminate our learning throughout the province and (if benefit is confirmed) lead widespread adoption of bedtime prescribing in patients that can be expected to benefit.

Design: Event-driven Prospective Randomized Open Blinded End-Point (PROBE) Study.¹³

Population: Community dwelling hypertensive Albertans who use BP medication.

Inclusion Criteria: 1) Hypertension diagnosis; 2) ≥ 1 BP medication; 3) Community Dwelling

Exclusion Criteria: 1) Palliative; 2) Unable to provide informed consent; 3) Personal history of glaucoma or use of glaucoma medications. *Glaucoma is an exclusion because nocturnal hypotension (i.e. excessively low blood pressure while sleeping) has been associated with ischemic optic neuropathy in such patients.*¹⁴⁻¹⁶

Intervention: Switching (*one at a time as tolerated*) one or more morning BP meds to BEDTIME (or *maintaining* bedtime BP meds if already taken at that time). All decisions related to which, and how many, medications to switch are at the discretion of the care provider.

Comparator: Maintaining (or switching to) morning BP medication use.

Outcomes: The SPOR Data Platform is partnering with us to provide the administrative claims data that define most of our outcomes. Other outcomes come from participant telephone interview. The SPOR Data Platform will monitor our primary outcomes quarterly during the trial, and the trial will conclude once 406 participants have experienced primary outcome events. Outcomes include:

Primary: Composite of all-cause death and hospital admission or emergency department visit for acute coronary syndrome/MI, heart failure, or stroke (validated coding algorithms for each¹⁷⁻¹⁹).

Secondary: 1) Each element of the primary outcome on its own; 2) All-cause hospital admission; 3) Resource intensity weight & length of stay; 4) Nursing home admission; 5) Self-reported Quality of Life (from EQ-5D: 5 functional health measures and patient's self-reported overall health score).

Safety: 1) New diagnosis of glaucoma; 2) Glaucoma surgery; 3) Hip fracture; 4) Self-reported worsening of vision (since study start - unchanged, slightly worse, much worse); 5) Self-reported Syncope/fainting (prior month - yes/no); 6) Self-reported falls (prior month - yes/no).

Tolerability: 1) Number of overnight urinations; 2) Self-reported nocturia burden (prior month - no problem, minor problem, major problem); 3) Light-headed or feeling "faint" (prior month - yes/no).

Process: 1) Proportion of baseline BP meds switched to bedtime at 6 months - broken down by antihypertensive class; 2) 24-hour Ambulatory BP at 6 months (see Sub-Studies section below).

Sample Size Estimate:

Number of primary outcome events to stop the study: We are planning a survival analysis with 1:1 allocation and want to detect a 25% relative risk reduction with probability (power) 0.8 and Type I error probability $\alpha=0.05$. This requires 379 events (conservatively assuming no useful covariates). We will increase this by 7% to **406 primary outcome events** in order to account for withdrawals, noncompliance and loss to follow-up (which combined are projected to be 5% in our pilot study).

Expected event rate: Utilizing Canadian administrative claims data for 2005/06^{1,2} we are able to predict (individually) the rates of mortality, and hospitalization for stroke, MI and CHF in a hypertensive cohort with the same age distribution as hypertensive Canadians (detailed calculations available on request). Although individually these estimates are robust, if we were to combine them together they would overestimate occurrence of our primary outcome because: 1) our survival analysis looks at time to first event and some of the predicted events will occur in the same individual; and 2) age specific rates of cardiovascular mortality and morbidity have been steadily falling since the 1970s and rates in 2016 are likely to be lower than those from 2006.¹ Countering this is our expectation that older (retired) patients are more likely to participate in a trial such as this. To avoid over counting in estimating our event rate we have used only hospitalization for stroke, CHF and MI (and excluded death) when powering. We did this believing death to be the most likely event to overlap the others. The 2006 to 2016 reduction in CVD risk and the potential greater proportion of retirees responding to our recruitment have opposite effects on event rates and should, at least partially, balance out. Altogether this gives us an **expected event rate in controls of 2.9%** and an expected event rate in the overall trial population of 2.0% (conservatively assuming MAPEC-like risk reduction to produce the smallest possible overall event rate).

Sample size / Study duration: We are budgeting to enrol 35 patients per day. Assuming steady enrolment (50 weeks/yr) until sample size is met, **8,750 subjects enrolled over 12 months and followed for another 22 months** is expected to provide the required 406 primary outcomes.

Recruitment / Patient Flow: Community family physicians (members of the Pragmatic Trials Collaborative) have already been recruited to participate in pragmatic trials like this by Drs. Garrison, Allan, Kolber, and Korownyk (see "Feasibility"). Patient flow steps: (see Appendix pg. 2)

1. Identifying Patients: Using billing / EMR software, providers produce a list of hypertensive patients and remove anyone they feel is palliative or incapable of informed consent.

2. **Informing Patients:** The study team gives providers one pre-stuffed envelope for every patient on their list. Provider offices affix patient address labels and mail the letters. Patient envelopes contain: 1) *a letter of introduction* from their physician; 2) *a study information sheet* with phone number reaching our study team; 3) *a consent form* with postage paid return envelope; and 4) *a "provider work sheet"* (see appendix page 1).
3. **Consenting Patients:** The study team dialogs with patients calling the contact number and obtains consent (online using REDCap or by return post) if interested and eligible. After consent 1) baseline characteristics not available in administrative data are collected such as smoking status, and frailty status according to the Tilburg Frailty Indicator²⁰⁻²², and 2) participants are asked to make (and advise us of) a study specific BP visit with their provider.
4. **Allocation:** Upon calling us back with the date of their provider appointment (demonstrating commitment), patients dialog directly with a research assistant with no clinical patient interaction who obtains their allocation to treatment or control (via a simple randomization – no stratification or blocking) from the central REDCap™ server's randomization module (which ensures irreversible and concealed allocation). Patients are also walked through 1) writing a list of current BP meds at the top of the provider worksheet and 2) circling the worksheet's allocated treatment arm (either "use BP meds in morning" or "use BP meds at bedtime").
5. **Intervention:** The patient arrives at their provider's office for the scheduled follow-up visit and hands them the provider worksheet. Providers observe the allocation (and the list of current BP meds), adjust medications as necessary and arrange clinical follow-up as per their judgment.
6. **Study Follow-up:** Post intervention, the study team calls patients at 1 week, 6 weeks, 6 months and every 6 months until study conclusion. After 6 months patients can opt for automated REDCap enabled on-line follow-up (chosen by 31% of patients in our pilot study), which improves study efficiency. These follow-up interactions supplement claims data to establish death and nursing home admission and collect data on adverse effects and patient experience.

Interim Analysis: An independent data safety monitoring board (DSMB) organized and chaired by Jim Wright (Cochrane Hypertension Review Group Coordinating Editor) will review all available data upon reaching 200 events (anticipated to occur at 20 months). If $p \leq 0.001$ for benefit, or $p \leq 0.05$ for harm, the committee will apply clinical judgement to breaking the trial early.²³

Sub-studies: 1) **Achieved BP:** 100 intervention and 100 control subjects will undergo 24-hour BP monitoring at 6 months to determine BP differences between groups. 2) **Diuretic tolerance:** Upon enrolment of 200 patients whose only BP med is a diuretic (including combination pills) we will analyse, in all patients allocated to bedtime, 6-week compliance to allocation for all BP meds. If diuretic compliance is substantially worse than other drugs our adaptive trial design²⁴ will exclude the enrolment of additional patients whose only BP medication is a diuretic.

Statistical Analysis: The primary outcome will be analysed by Cox Proportional Hazards Survival Analysis using baseline covariates which include: age ≥ 80 , sex, frailty (binary - from the Tilburg Frailty Indicator²⁰⁻²²), cognitive impairment (binary - from the Short Blessed Test²⁵⁻²⁷), smoking status, hospitalization in last 6 months, ≥ 3 baseline BP meds (resistant hypertension), presence/absence of specific diagnoses (diabetes, CHF, stroke/TIA, coronary artery disease), and significant renal impairment (a claims data renal insufficiency diagnosis, eGFR less than 40, or dialysis).

Feasibility of Proposed Activities:

1. **Physician Enrolment:** Our trial network, known as the Pragmatic Trials Collaborative (www.PragmaticTrials.ca) and led by Dr. Garrison (our PI), recently completed enrolment of its first RCT (successfully piloting BedMed's recruitment methods - albeit with a different question). This trial (INRange³²), which examines the effect of the timing of warfarin ingestion on time in therapeutic INR range, had a remarkable 236 British Columbia (BC) and Alberta family physicians from 53 different communities mailing recruitment letters (78.4% in Alberta). **This degree of physician engagement is unprecedented.** To our knowledge, the largest published non-industry RCT relying on Canadian FPs to recruit patients involved 79 physicians.³³ INRange engaged triple that number. Our membership grew by 131 physicians last year and now numbers close to 400 across both provinces. And we expect higher

participation in BedMed - as this trial holds greater interest for clinicians than our pilot. Dr. Allan also co-hosts a popular medical podcast (Best Science Medicine) through which FP enrolment can be enhanced and the AMA's Primary Care Network Program Management Office (PCN PMO) has agreed to assist physician recruitment using 3 different newsletters that reach all PCN physicians. We are well positioned to recruit for this trial. Additionally, we will work with the *Alberta Strategy for Patient Oriented Research – Primary Integrated Health Care Innovation Network* (SPOR-PIHCIN), to use the hoped for PRIHS3 grant as matching funds for an upcoming CIHR Phase III SPOR competition. As we already have a functioning network in BC, and as Dr. Alex Singer, the Manitoba CPCSSN lead (Canadian Primary Care Sentinel Surveillance Network) has approached us about running BedMed in Manitoba CPCSSN practices, we are well positioned to expand to other provinces.

- 2. Patient Enrolment:** Based on an evaluation of six typical fee-for-service family practices, we estimate the average family practice to contain roughly 250 hypertensive patients. This is consistent with the average size of a FP practice (1500 - 1800 patients) and the prevalence of hypertension in adult Albertans (19.6%). In INRange, 20.4% of patients receiving letters from their family physician called to enroll in the trial. BedMed patients will be younger (on average) than those in INRange and may be less likely to respond to a letter from their doctor. Patients already taking BP meds at bedtime may also be disinclined to participate. Hence, although a similar response is possible, we will conservatively plan for only 12% patient response. We plan to engage **365 physicians** (a number that is already within the reach of our membership), who will send letters to 250 patients each. We expect 85% to use BP meds, and 12% to be interested and eligible. Thus **we expect to enroll 8,750 patients** from 91,250 mailed letters.
- 3. Study Duration:** We are budgeting to enroll 35 patients per workday (with 50% of the working hours of 3 research assistants dedicated to telephone enrolment). At this anticipated recruitment rate we will enroll all 8,750 patients over 1 year. This would reach our target # of primary outcomes to stop the trial 2 years and 43 weeks from the start of enrolment. However this is a conservative estimate based on the lowest possible (MAPEC-like risk reduction) event rate. If instead the 25% relative risk reduction we powered for is found, events will accumulate much more quickly and the trial will end 6 months sooner (at 2 years 4 months). If the benefit exceeds a 25% relative risk reduction, it is also possible that the trial will terminate for benefit at the interim analysis (if the true benefit is a 37% reduction in events there is a 50% probability that a p of ≤ 0.001 will be observed at the 20 month DSMB review). Also, if we are successful in the SPOR PIHCIN competition, we would expect higher (and faster) enrolment and hence a shorter wait to reach the event target which determines trial duration. We expect the trial to complete, and the broader province-wide implementation phase to begin (allowing 2-3 months for analysis), at some point 23 to 36 months after the start of enrolment.

Supporters: This project's key commitment comes from: 1) the volunteer primary care providers of the Pragmatic Trials Collaborative, whose lead is PI of this application (Garrison); 2) the Alberta SPOR Data Platform, whose Lead is a collaborator (McAlister); 3) the U of A EBM Knowledge Translation Team (Allan - co-PI, Korownyk, Kolber - collaborators); 4) the Health Quality Council of Alberta, whose Director of Health System Analytics is a collaborator (Lahtinen); and 5) the Alberta PCNs, connected through another collaborator (Lewanczuk) who is AHS Senior Medical Director for Primary and Community Care. We have also obtained support from 3 Scientific Clinical Networks (SCNs), the Alberta College of Family Physicians, the Rural Physician Action Plan, the AMA, the SPOR Pragmatic Clinical Trial Platform, the SPOR PIHCIN applicant group, the Departments of Family Medicine at the University of Alberta and University of Calgary, and the Cochrane Hypertension Review Group (who will serve as Data Safety Monitoring Board and provided peer review for our ethics application).

Data Requirements: The Alberta SPOR Data Platform is partnering with us on the BedMed Initiative and will provide all data extractions and required linkages with administrative holdings of AHS DIMR (specific to our participants) including: 1) the Discharge Abstract Database (information on all Alberta acute care hospitalizations - specifically discharge diagnoses, procedures, discharge disposition, resource intensity weight, and length of stay); 2) the National Ambulatory Care Database (information on all Alberta ER visits – specifically emergency department visit

diagnoses); 3) the Pharmaceutical Information Network Dispenses (all prescriptions dispensed by participating community pharmacies - to be used in the cost analysis); and 4) outpatient laboratory data (which, together with Netcare laboratory data, will provide information on the creatinine/eGFR lab tests used to define renal insufficiency). All participants in our pragmatic trial will have provided written informed consent for these records to be accessed.

In addition to the complete set of outcomes and baseline characteristics provided for the 20-month interim analysis and for the full study, we will also receive from the SPOR Data Platform quarterly reports on the occurrence of our primary outcome (confirmed in the attached letter of support from Dr. McAlister, the Alberta SPOR Data Platform Lead) so that we will know when to end the study.

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Log of BedMed Amendments

Date of Submission (D-M-Y)	Date of Acceptance (D-M-Y)	Summary of the Amendment (Wording as submitted to the ethics board)
08-01-2017	03-02-2017	There are no substantive changes to our methodology. However, in the course of writing another related grant application (and receiving further feedback on study refinements from co-investigators in the process), and having multiple patient working group meetings to discuss the study documents and the data to be collected, there are numerous minor changes to the wording of all our study documents (patient information, provider worksheet, and consent form) and to the specific data collected directly from participants during interviews. The patient consent form has changed - not because of anything new that we are doing, but because of patient feedback. Specifically, it now explicitly includes the statement: "I understand my PHN number will be used to link my data to hospital, emergency room, physician office visits, pharmacy and lab data." There is also a change in the number of subjects that will undergo 24 hour BP monitoring (as a result of a new sample size calculation for that sub-study). We are also taking the opportunity this amendment presents to update some previously unknown information such as the study registration and funding speed code (successful PRIHS grant). Additionally, there is a new sub-study described in which we are comparing the characteristics of the practice populations from which the study subjects were drawn (i.e. everyone in the same participating practices with a diagnosis of hypertension and one or more BP lowering medications) to the same characteristics of the participants themselves. In so doing we hope to learn how representative our participants were of the general population. This will help with assessing external validity. The SPOR data platform will perform this analysis, and only provide to us aggregated (i.e. non-identifiable) information surrounding the practice populations to which we will compare our study participants. Specifically we will look at both populations according to our covariates including gender, age, hospitalization in the prior 6 months, proportion with specific conditions (diabetes, CHF, stroke/TIA, coronary artery disease, sleep apnea), and proportion with ≥ 3 BP lowering meds (resistant hypertension), ≥ 4 non BP lowering medications (polypharmacy), and kidney disease. We will also look at the two populations according to comorbidity, rural vs urban, high attachment vs low attachment, and the practice size from which they were recruited. No individual physicians will be identified - information will only be provided to the researchers in aggregate (pooling data from approximately 365 participating physician practices).
07-02-2017	13-02-2017	Adding in the use of the consent for Section 34 compliance of HIA: Consent to Disclose Health / Registration Information. This will be provided as a paper consent form and will be available as an electronic survey provided participants have their own 2-step login created at the beginning of the

		study to verify themselves. This will be used in the event that we need to verify information about them or their disposition at the end of the study. Participants can decline this consent but can still participate in the study without it.
20-02-2017	08-03-2017	7.1.1.0: Correction of a typo on Page 5 in Participant Information Package - Alberta, dated 08 Feb 2017 (3 digit participant code -> 5 digit) 7.1.5.0: PDF documentation of Interview Questions from REDCap, dated 17 Feb 2017 (Including changes to Screening, Baseline, Randomization, and Week 1 Follow-up)
30-03-2017	04-04-2017	7.1.1.0: Patient-facing poster for display in study-affiliated family physician offices for patient engagement, dated 31 Mar 2017 7.1.5.0: Zip file containing PDF documentation of Interview Questions from REDCap (Including changes to Screening, Baseline, Randomization, Week 1 Follow-up, Week 6 Follow-up, and Month 6 Follow-up), dated 31 Mar 2017
03-05-2017 07	11-05-2017	Adding brief questions at the Week 6 time point to gather participant feedback regarding study design and methods. These questions are not necessary for study participation and participants can opt to not participate.
29-05-2017	01-06-2017	Based off feedback from participants and clinics, we shortened reformatted the information package to increase readability. Furthermore, the letter of introduction from the family physician has been simplified and revised to increase clarity.
27-06-2017	07-07-2017	This protocol amendment opens up study enrolment to residents of BC and Manitoba. Study-wide sample size is expanding from 8,750 in the original protocol to 11,700. The enrolment process has also been expanded to allow for word-of-mouth recruitment from patients not belonging to Pragmatic Trials Collaborative-associated physicians. Inclusion/exclusion criteria have also been clarified and edited to ensure patient safety. Patient study flow has been optimized in accordance with feedback from both patients and providers in order to reduce study burden.
03-08-2017	17-08-2017	Most up to date Month 6 interview questions attached (dated 03 Aug 2017). Also included in this application are slip-in notes for a lottery incentive for patient recruitment purposes.
16-11-2017	29-11-2017	An updated Month 6 Survey has been created on RedCap. Documentation is attached. In the Screening instrument, physician information is now requested before assessing for inclusion criteria rather than after. Physician information includes the doctor's first and last name as well as any additional identifiers (eg. clinic name, clinic location or phone number). These database changes allow us to account for the different information packages individuals may have received from their physician and subsequently, allows us to follow the study processes accordingly. A feedback slip is also being submitted, to be mailed along with the information package to potential participants. The purpose of this slip is to understand some of the reasons why individuals choose not to participate. The feedback from individuals who have declined to participate may allow

		us to improve our methods to increase public interest. Individuals will be able to provide feedback through three different options. They may call us at 1-844-492-7570, or email us at bedmed@ualberta.ca . If individuals wish to be anonymous, they can visit the link: http://is.gd/albertafeedback . This will direct them to an online survey through RedCap.
01-12-2017	20-12-2017	Uploaded for approval is a consent form for family physicians that are part of the Pragmatic Trials Collaborative. This consent is to allow the Analytics, Performance, and Reporting Branch of Alberta Health to query provincial billing data on behalf of family physicians in order to optimize the list of qualifying patients and reduce burden and workload for clinics. This data would then be provided back to the physician, and their staff would produce the mail out as per usual. The study team will NOT have access to the data at any time during this process - such data remains only between Alberta Health and authorized staff at the physician's clinic. Also included is an SOP describing tiered participation and withdrawal in order to formalize the manner in which patients can continue to be part of the study in a limited fashion (for example, data being collected via databases rather than through direct interactions).
02-02-2018	09-02-2018	An optional 24-hour Blood Pressure monitoring sub-study will be conducted to assess blood pressure differences in the morning and bedtime allocation groups. 151 intervention and 151 control participants will wear an ambulatory BP monitor for 24 hours. Participants residing near an Advanced Care Network (ARC) location who are completing their month 6 follow-up survey online are eligible to participate. Upon completion of the survey, participants will be invited to take part in the optional sub-study and provided with an information package to read online. If they are interested, the study team will contact them within 1 business day, or alternatively, participants may also reach our study team on the number listed on the information package. The study team will describe the sub-study and complete the online consent. A referral will be sent to ARC and within 72 hours, ARC will contact eligible participants to schedule an appointment for the fitting of a blood pressure monitor. Participants will complete the 24 hour blood pressure monitoring and return the device to the ARC clinic. The study team will receive the data and enter it into RedCap. Participants have the option of having this data sent to their physician if they consent to this sharing of information. In such cases, data will be faxed to the participant's physician with a cover page as required for health data transmission.
08-03-2018	27-03-2018	We have been enlisted Habit Creations to revise our patient-facing documents and create a more user-friendly and engaging design where possible. In this amendment, we submit visually revised consent forms and information package. The consent forms do not contain any wording revisions, simply they visually match the study brand that has been created. The information package will not display entirely on your screen - the biggest format change will be that is all contained on a single 'tabloid-sized' paper that is folded and can be held like a magazine. There are wording revisions to this document that are further highlighted to show where a word or phrase was added/edited/moved/removed. These changes are being submitted for review and approval and then based on the final approved, a public-facing campaign will be created that visually matches.

		In order to increase recruitment, we have contracted an advertisement and marketing consultant (Habit), to produce a public-facing campaign. This advertisement campaign takes the form of a website landing page, as well as various forms of advertisement. The advertisements (described below) will bring interested individuals to the study website, where detailed information based on the approved Participant Information Package will be displayed. Patients can download the actual package on the website. We are also allowing participants to self-screen using the established inclusion/exclusion criteria in order to quickly determine if they are eligible for the study, and to provide contact information for contact by study staff if eligible. Study contact details are also available in case patients are interested in contacting the study team directly. Attached to this amendment are links to the video ads to be used - we would like to be able to create further variations of these videos based on approved content as needed to increase ad relevance based upon analytics. These videos will be used in both online campaigns (Facebook and YouTube) as well as in clinics across Alberta. We will be working with AHS and the AMA to display ads at these locations. Online advertising allows us to make adjustments and refine our message over the course of the campaign. We can also focus on our audience in a much more targeted way than through other tactics. For this campaign, all advertising will be geotargeted within Alberta. We've selected Facebook, YouTube, and Google Search as our initial channels for online advertising. This will be expanded to other platforms such as Twitter, Instagram, or other social media using the same videos and images, adapted for the platforms as needed. Facebook 80% of people aged 45–54 are on Facebook, with 74% of those aged 55–64 having membership on the platform and a drop-off to about 58% of those over 65. Statistics tell us that 89% of Canadian Facebook users read, view, or search for information on Facebook at least once a week, with 68% of them checking at least daily. On Facebook, we'll target men and women aged 50+ based on their interests and liked pages. Ad mockups have been attached to this amendment. Google Search 72% of people turn to Google to search for answers to specific problems. We'll develop search ads based on the following keyword themes: hypertension, high blood pressure, volunteer, and medication timing for high blood pressure. These keyword themes were selected based on monthly search volumes, ad auction ranking opportunity, and relevancy to our content. Wording of the advertisement comes from the study website, based upon keywords used by the patient. YouTube 22% of Canadians watch YouTube daily, 47% watch weekly, and 67% watch monthly. We'll use YouTube pre-roll video advertising to reach males and females. We'll focus on channels related to health, fitness, and family, as well as select popular interest channels specific to people aged 50 and older.
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		As a result of the ad campaign, some participants calling us may not have a family physician available to guide them through the medication timing change process (if required). In order to accommodate these participants, we have created a set of guidelines that can allow some participants to self-change their medication timing if they are medically simple. These guidelines were created in consultation with our collaborator Dr. Raj Padwal (Director of the Hypertension Clinic at the University of Alberta and the past Chair of the Hypertension Canada CHEP Guidelines Task Force) and Dr. James Wright (Hypertension Specialist and Coordinating Editor of the Cochrane Collaboration's Hypertension Review Group). Dr. Wright's letter of support for our proposed method of guiding medically simple patients to make their own timing change is attached. Medically complex participants without a family physician can also participate in the study, but such patients will be guided through the process over the phone by a BedMed associated physician or pharmacist. This medication self-timing change is available to all participants in the study if they are medically simple. This includes participants who do not want to visit their family physician and participants whose family physicians have asked as to provide this service to patients deemed medically simple. Participants can always opt for a physician-guided change if they prefer to do so, even if they are deemed medically simple. If a participant elects for a self-timing change, RAs will use the established algorithm to change the timing of up to 2 blood pressure medications, one at a time as tolerated, with at least a month between changes and a follow-up telephone call one week after each change. If any adverse effects are reported, a BedMed physician or pharmacist will speak directly with the participant. RAs will not change the timing of Alpha Blockers as these drugs are more often prescribed for benign prostatic hypertrophy than to lower blood pressure. Medically simple patients, for the purpose of this protocol, must 1) state to the RA that they have never had any heart related issues of any kind, 2) must not use Tiazac XC (which is a delayed release medication whose timing is more complex), 3) must not use furosemide (which is often in use for reasons other than hypertension), 4) must not use BID antihypertensives (whose timing may be more difficult to change), and 5) must use no more than 2 antihypertensives, excluding alpha blockers (to avoid working with patients who have resistant hypertension). If patients are medically simple and comfortable being guided to make their own timing change the RA will switch medications in the order ABCD (ACEI/ARB, beta-blocker, calcium channel blocker, diuretic). The process may be visualized in the submitted BedMed Patient Flow & Medication Change Process document. An algorithm embedded in the REDCap Database will check the criteria for each patient and display a "go or no-go" message for self-change eligibility. The system will also guide the actual process for the RA in a step-by-step manner. (Screenshots attached in Documentation) Also included are the screenshots of the Month 12 Interview Questions. These are a repeat of the Week 6 questions already approved.
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22-02-2019	18-04-2019	BACKGROUND Increasing the proportion of patients responding to recruitment packages is important to the success of any trial and the BedMed team has attempted a variety of changes in recruitment materials to accomplish this. One novel change that has not yet been attempted is to provide, in the physician's letter of introduction, an up to date number of people who are already participating in the trial. Research in other areas suggests that knowing other people are participating increases the participation rate. We would like to see if that would hold true for this type of recruitment as well. WHAT'S NEW We wish to create a second version of the physician letter of introduction that has 1) a single extra sentence, and 2) a one word addition (the word "too") in the sentence that follows. The newly introduced sentence and the altered sentence which follow become: "This study already has over 1,700 Canadians with high blood pressure taking part. If you too choose to participate, our office will assist you in making and monitoring any medication changes that might be required." Our intention is to create equal numbers of the original physicians letter (which doesn't state the number of participants), and this new version of the physicians letter (which does), and to provide every physician office with both versions randomly distributed in the bundles of recruitment envelopes we ship to them. The only other change will be that the date of the letter will be even (e.g. 12th, 14th, 16th) if it is the original version, and odd (e.g. 13th, 15th, 17th) if it is the new version. When interested patients read the material and call us about participating, the research assistants will ask for the date on the letter and use that to determine which version of the letter they are responding to. This robust method of evaluating differences in response rate provides us the most reliable way of assessing if the alteration in the letter increases enrolment. We are taking this step because we believe this knowledge, on its own, would be worth sharing with other research groups interested in improving patient participation rates. We anticipate having roughly 6400 (or more) recruitment packages mailed before enrolment is complete. With a baseline 8% response rate, we anticipate having enough power to detect a 2% absolute difference in response rate to an alpha of 0.05 with 80% power.
08-04-2020	16-04-2020	We already described, and received approval for, a sub-study where we are looking to see how representative clinical trial participants are of the people who were invited to participate in the trial. Specifically, we received approval to obtain from AHS aggregate results of the demographic and comorbidity characteristics of all patients attached to participating practices that we are already collecting on BedMed participants. In so doing we can compare the two groups and see if clinical trial participants are different from the population from which they are drawn. In this amendment we are requesting to do 2 additional things:

		1) Have AHS also provide us aggregate results of the same characteristics for patients that appear eligible for our trial within the whole province (to see how representative the participating GP practice patients are of all eligible patients). and 2) Add, for all 3 of these groups (trial participants, eligible patients attached to participating GPs, and all eligible Albertans) two further groups of variables that would help us understand how these patients may be different. For all three groups we would only learn these results in aggregate, and be unable to attach the characteristic to any given individual. These two groups of characteristics are A) PRIOR 3-YEARS SCREENING TESTS (including performance of a PAP smear, FIT test, colonoscopy, PSA test, and mammogram), and B) PRIOR 3-YEARS PREVENTIVE THERAPIES (including dispensing of a shingles vaccine, statin, and osteoporosis medication). We will also be reporting (from aggregate data only) the rate of all-cause hospitalization in each of the three cohorts during the study period.
13-05-2020	19-05-2020	We have already described an analysis of who participates in clinical trials, as compared to who is invited. In this analysis we are looking at a variety of characteristics and comparing them in those who are participating in the trial, to those who are eligible and attached to participating practices (who should have been invited to participate), and to those who are eligible in the province at large. Those characteristics are things like co-morbidities, screening behaviours, and preventive therapies. In this amendment, we are requesting to expand the data we are collecting to include the number of visits with specialists, and the number of visits with family physicians, in the year prior to enrolment (or the year prior to the study start for those who are not enrolled in the study). We also wish to create 2 new cohorts for the same analysis. These cohorts include 1) those who were lost to follow-up or dropped out of the study (and did not withdraw consent for us to access their data), and 2) those who were unable to comply with the intervention timing change. We will only obtain this data in aggregate and hence will be unable to identify these characteristics in any specific individual. I have changed section 2.1 5.0 (on sub-studies) to include: AMENDMENT May 13 2020. We have expanded the data collected in this sub-study to include the number of specialist visits in the preceding year, and the number of family physician visits in the preceding year. We have also added 2 additional cohorts in which to examine the same characteristics. This includes 1) Those who were lost to follow-up or dropped out of the study (and did not withdraw consent for us to access their administrative claims data), and 2) Those who were unable to comply with the timing change.
22-10-2020	29-10-2020	Study Coordinator name change, contact information remains the same.
21-05-2021	31-05-2021	Contained within the existing approved ethics application is a sub-study referred to as the "External Validity" sub-study. In this sub-study, using only administrative claims data, we look to see how representative our trial participants are of those who appeared eligible for the study within 1) the entire Alberta population and 2) within the patient panels of the

		participating family physicians. This involves Alberta Health Services (AHS) creating these two cohorts for us to compare to. All of this has already been approved. In order to build the cohort of eligible patients attached to participating practices (of which there will be around 70,000), the AHS analyst carrying out the work needs to use the family physician "PracIDs" to identify their eligible patients. The PracID is created by Alberta Health to identify physicians within the administrative claims data. The problem for which this amendment is being generated is that PracID is one of the few things AHS does not have access to in the administrative claims data without explicit approval from Alberta Health to use it. Alberta Health is willing to give this approval but they have requested us to do 2 things first. 1) They want us to send an e-mail to all the physician participants asking them to let us know if they object to AHS using their PracID – i.e. to let us know if they want to opt out of that information sharing, with the default being that the PracID will be shared if no objection is voiced. 2) They have asked us to ensure that we have a waiver of consent for this sub-study from the ethics board. i.e. That we do not need to obtain consent from individuals within the two administrative claims data derived comparison cohorts, nor do we need to obtain specific consent for the sharing of the PracID from the physicians who are already participating in the study. To be clear, we are not changing anything in the sub-study other than specifying that Alberta Health will permit Alberta Health Services access to the PracIDs of the participating physicians and that we will let the physicians know this is happening, with an opportunity to opt out if they don't want it to occur. Relevant to this, is that the analysis is carried out entirely by the data steward, that researchers never see patient level data, and that the smallest of the two comparison cohorts (eligible patients attached to participating practices) will have around 70,000 individuals contributing data. We are not collecting or reporting any information about participating physicians. Merely using the PracID to create the cohort of attached eligible patients. Also relevant is that, we believe, the participating physicians would have assumed, as we did, that AHS already had access to this information. We have been told by AH that their intention is for the AHS-AH information sharing agreement to be updated to include AHS having access to physician PracID, but this will not occur in the time frame of our study. Within the ethics application, for the purpose of this amendment, we made a single addition at the end of: 2.1 Study Objectives and Design 5.0 For clinical trials, describe any sub-studies associated with this protocol Which reads: AMENDMENT May 21 2021. To create the previously described cohort of eligible individuals attached to participating practices it is necessary for
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		Alberta Health to share each participating physician's PracID with Alberta Health Services, who is conducting the analysis. At Alberta Health's request, we will be notifying participating clinics that this is occurring, by e-mail or by telephone, in order to give individual physicians the opportunity to opt out if they don't want Alberta Health Services to be able to use their PracID for the analysis.
23-08-2021	25-08-2021	For participants who experienced an event that was reported to BedMed team during a 6 month follow up. All reported events will be reconciled with administrative claims data, and any events unaccounted for will be followed up by fax with participants physician to confirm the event. Fax is the typical form of communication with physician when communicating about participant/patients.
01-03-2022	04-03-2022	Additional questions for all participants, and close out survey for participants who end their participation in the study, or complete the study.
24-03-2022	31-03-2022	The project recently received an extension in funding, for those participants who complete their surveys online are unaware of the changes as we do not speak directly with them. The survey intro will be amended to inform participants of the new extended study timelines. For study participants, the extension means there will be additional surveys to complete (at 6 month intervals) until the amended end date (an additional year). Due to the nature of how the study is set up in REDCAP, the various provinces and subsequent differences in approvals dates, and study participants starting and ending the study at different times we are unable to state how many additional surveys each specific participant will have as all participants see the same information when they complete a survey. By providing the end date (End of 2023) we feel that makes it clear what to expect for participants regarding the additional time in the study. In addition, the study phone number/email are provided and an opportunity to end involvement in the study "Are you still interested in participating in the study (Y/N)" in each survey, provides an opportunity for participants to discontinue their participation or talk with the study team.
02-08-2024	02-08-2024	This amendment is intended to describe how we would share de-identified patient-level data with other researchers. Sharing patient level data has become the norm for RCTs and all the big journals require us to have a statement about how other researchers can request our de-identified patient level data. We would like to be as open with sharing our patient level data as possible. The following are what we believe you need to know about our data sharing plans. 1) PARTICIPANTS WERE INFORMED ABOUT OUR DATA SHARING PLANS WHEN THEY ENROLLED IN THE TRIAL: Prior to enrolment, during the consent process, we advised all participants that we would be sharing deidentified data from the trial. Specifically, the material accompanying the consent stated: 'To maximize the value of this study after our analysis is complete, we will make our raw data available over the Internet (with all identifying information removed) so that other research groups can verify our findings and explore questions of their own.' This statement can be found in the very first document attached in the Recruitment Materials section of this ethics application with the title "Participant Information

		Package - Alberta 22Oct 2020 (0.07)" The relevant text is on page 3 in the last paragraph of the section entitled "Study Confidentiality". 2) WHAT WE WILL SHARE IS COMPLETELY DE-IDENTIFIED: The dataset to be shared is an Excel file of our analytic dataset with variables that could be identifying removed. Specifically, we have removed 1) Birth date (including only age, in years, at time of randomization), 2) Study ID, 3) Province of residence, 4) all event dates (instead providing a “days since randomization” for each event), 5) all dates related to the last contact with the participant – instead using a number of days post randomization at which censoring occurred and providing the type of censoring. There will be 3,357 individuals in this dataset, making it near impossible for anyone to be identified. We will also remove baseline dementia diagnosis, as a precaution, as this is potentially sensitive information. There will be a second page of the Excel spreadsheet that will have data from a smaller number of individuals (302) who went for 24-hour BP assessment. However, this subgroup data will not be linked to the other spreadsheet data, and we will present only BP numbers and variables derived from these numbers. Hence knowing someone underwent 24-hour BP monitoring would not help in identifying their data. Alberta Health Services legal representatives were advised of our data sharing plans, and our steps to de-identify the data, and they have approved it. 3) WE WILL SHARE VERY OPENLY, BUT WE WILL TRACK WHO IS DOWNLOADING OUR DATA, AND WE WILL ASK THEM TO ACKNOWLEDGE THAT THEY WILL NOT ATTEMPT TO IDENTIFY ANYONE IN THE DATASET: Specifically, we will: a) Create a new UofA hosted REDCap database to track who is downloading our data and for what purpose. b) Provide a link on the BedMed page of the Pragmatic Trials Collaborative website to a REDCap survey that asks for: i) Name of the individual responsible for handling the security of the analytic dataset upon receipt (the “Data recipient”). ii) Name of data recipient’s affiliated academic institution. iii) Country of data recipient’s affiliated academic institution. iv) City of data recipient’s affiliated academic institution. v) Data recipient’s academic status (mutually exclusive radio buttons for Professor, Associate Professor, Assistant Professor, Clinical faculty, non-academic) vi) Description of the use to which the data will be put. vii) E-mail address of the data recipient c) Once that e-mail address is entered we will use the e-mail address to send the data recipient a REDCap consent waiver asking them to agree that they will: i) Agree not to share the data outside of their immediate research team (i.e. anyone authorized to use the data must register with us themselves) ii) Agree not to attempt to identify anyone in the dataset iii) Agree to acknowledge the Alberta SPOR Support Unit for their curation of the data, and acknowledge Alberta Innovates and the Canadian Institutes of Health Research for the funding of the original work iv) Agree to offer one or more BedMed co-investigators the opportunity to participate as a co-investigator and/or co-author on any project whose findings are based, in whole or in part, on BedMed data.
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		d) At this point, if both points are agreed to, an e-mail is sent to Dr. Garrison with the request for data sharing. He will make the final decision on data sharing, and he will e-mail the analytic dataset directly to the data recipient if data sharing is agreed to.
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